# C2 · Predicting extreme hospital stays at admission

Business Intelligence IIB423T-1 · UDD · Instructor: Tomás Fontecilla · GROUP_01: Matías López, Cristóbal Ramírez · 08-11-2026 · Route: classification

## 1. Question and task

Beds are the scarce resource of public hospitals, and W1-W2 showed that extreme stays (longer than the cut-off of their GRD) are about 6% of stays but about a quarter of bed-days. **Question:** on the day of admission, can admission-time information flag the stays that will become extreme, so that discharge planning starts early? **One row** = one GRD-coded inpatient stay in the FONASA public GRD files (discharges 2023-2024, 72 hospitals). **Target** = 1 if length of stay (LOS) > cut-off of its GRD (Q3 + 1.5·IQR, base-GRD fallback below 30 cases); **positive class = extreme stay**. **Inputs** (known at admission): age, age band, sex, insurance, admission type, origin, admitting unit, hospital, admission month and weekday, transfer-in flag.

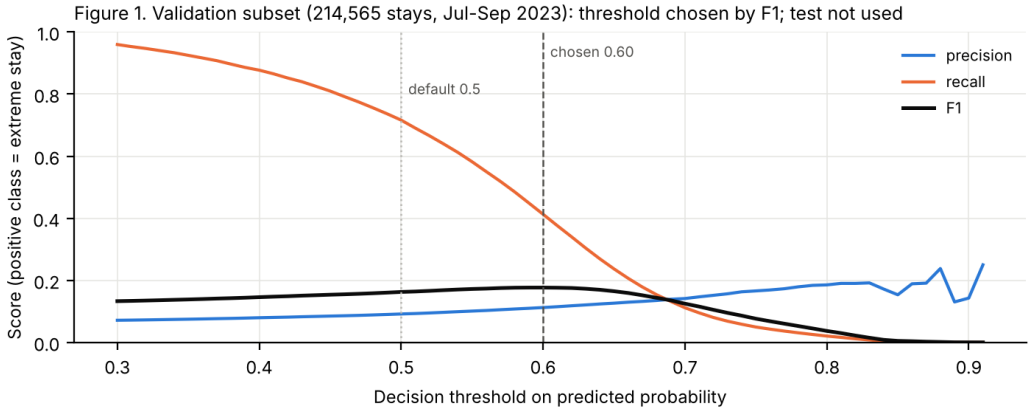
**W2 feedback applied.** (1) Diagnoses, GRD, weight, severity, procedures and every discharge field are excluded from the inputs; an assertion in the notebook stops the run if any of them enters. (2) The extreme-stay cut-offs are estimated on training admissions only, frozen, and applied unchanged to validation and test; they match the W2 table value by value (783 GRD codes) and a fingerprint is re-checked after each transform. The GRD is used only to build the target, never as an input.

## 2. Data, split and method

| Subset | Admissions | Stays (with label) | Use |
|---|---|---|---|
| Train | Jan-Jun 2023, discharged before 01-01-2024 | 421,352 (421,226) | Fit cut-offs, imputation, encoders, model |
| Validation | Jul-Sep 2023, discharged before 01-01-2024 | 214,640 (214,565) | Choose the decision threshold |
| Test | Jan-Sep 2024 | 655,321 (655,093) | One final evaluation |

The W2 chronological split is reused unchanged (Oct-Dec 2023 embargo; Oct-Dec 2024 right-censored; 2022 admissions left-truncated). Rows whose GRD has no frozen cut-off have no label and are not evaluated (228 in test). **Model:** logistic regression (scikit-learn), one-hot encoding plus scaled age fitted inside a pipeline on train only, `class_weight='balanced'` because positives are 6.3% of training stays, L2 penalty with C = 1 (default), no tuning. It is simple, fast on 421,000 rows and its coefficients read as odds ratios. **Baseline:** “flag every emergency admission”, a rule a bed manager can apply today and the strongest admission signal found in W2. **Threshold:** chosen on validation as the value that maximises F1 (0.60); the default 0.5 flags half of all stays. Test was used once, after model, inputs and threshold were fixed.

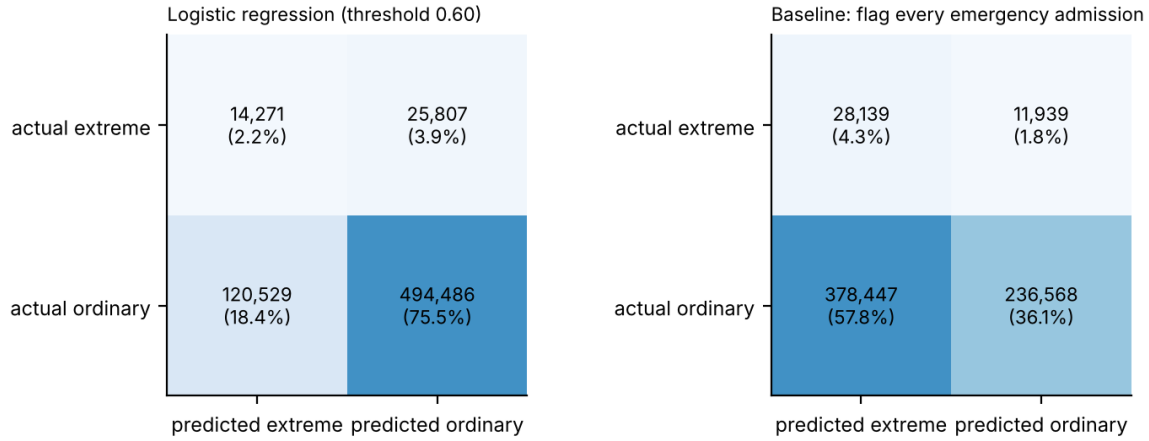
Figure 1. Validation subset (214,565 stays, Jul-Sep 2023): threshold chosen by F1; test not used




## 3. Results on the test set

| Rule (test, 655,093 stays, prevalence 6.12%) | Flagged | Precision | Recall | F1 | Accuracy |
|---|---|---|---|---|---|
| model, threshold 0.60 | 20.6% | 0.106 | 0.356 | 0.163 | 0.777 |
| baseline: flag every emergency admission | 62.1% | 0.069 | 0.702 | 0.126 | 0.404 |
| trivial: flag nobody | 0.0% | 0.000 | 0.000 | 0.000 | 0.939 |

Figure 2. Test confusion matrices, 655,093 stays admitted Jan-Sep 2024 (positive class = extreme stay)




Validation F1 was 0.177 for the model and 0.132 for the baseline, close to the test values, so the result holds from 2023 to 2024 admissions.

## 4. Interpretation

**What the model adds.** It flags 20.6% of admissions and catches 35.6% of extreme stays; 10.6% of its flags are real extreme stays, 1.7 times the base rate. The emergency rule catches more (70.2%) but flags 62.1% of admissions, and only 6.9% of its flags are extreme, barely above the base rate. The model has the higher F1 (0.163 vs 0.126), needs a third of the flags, and also covers obstetric admissions, which the rule misses by construction.

**Why accuracy is not enough.** “Flag nobody” has the highest accuracy (93.9%) and finds no extreme stay: with 6% positives, accuracy rewards ignoring the class we care about.

**Errors.** A false negative (25,807) is an extreme stay not flagged: planning starts late and the bed stays blocked; this is the costlier error. A false positive (120,529) is an ordinary stay reviewed unnecessarily: it costs staff time, not patient harm, but at 50% flagged (threshold 0.5) a discharge team would be overloaded.

**Main limit.** Admission fields carry weak signal: precision is 11% at the chosen threshold and stays below about 25% even when recall falls close to zero. Most causes of extreme stays (complications, social discharge barriers, waiting for transfer) appear after admission. The extract has no patient identifier, so repeated patients across subsets cannot be checked, and test lacks Oct-Dec 2024. A useful flag does not prove that earlier discharge planning shortens stays; that needs a trial or before/after study.

**Possible improvement.** Re-score each patient at 48-72 hours, adding information that exists by then but before discharge (for example, transfer to intensive or intermediate care, theatre use), and validate it with the same chronological split; set thresholds per hospital to match each team's capacity.

**Conclusion.** As a first screen, the model concentrates discharge-planning effort better than the emergency-admission rule, but it is not accurate enough to decide on its own.

## References

FONASA (2024, 2025). Bases de datos GRD públicas 2023 y 2024 (GRD_PUBLICO_2023.zip, GRD_PUBLICO_2024.zip), Data Catalogue of the dashboard PropuestaTableroGRD, public.tableau.com. MINSAL (2026). Establecimientos de salud, datos.gob.cl. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, 2825-2830. GROUP_01 (2026). W1, C1 and W2 submissions, github.com/Tobal2806/BI_Proyect. Notebook: GROUP_01_C2/notebooks/C2_model.ipynb (all numbers in this brief come from its saved outputs).